

Andrew Liu

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EDUCATION

Vanderbilt University

Expected: May 2026

Bachelor of Science in Computer Science, Applied Mathematics. GPA: 3.66

Nashville, TN

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures, Operating Systems, Object-Oriented Programming, Software Design, Algorithms, Linear Algebra, Differential Equations, Multivariable Calculus

EXPERIENCE

Vanderbilt University

Aug 2025 - Dec 2025

Undergraduate Course Assistant - Operating Systems

Nashville, TN

- Hosted weekly office hours for a class of 150+ students, providing debugging assistance for assignments in C, including memory management, concurrency, and process scheduling issues.
- Graded 30+ programming assignments per week, identifying bugs, memory leaks and other flaws to provide actionable technical feedback

Institute for Software Integrated Systems

May 2025 – Aug 2025

Software Engineer Intern

Nashville, TN

- Architected Vocab Lab, a cross-platform mobile vocabulary learning application using **Expo** and **React Native**, implementing the FSRS spaced repetition algorithm to optimize long-term retention with 90% target accuracy
- Engineered adaptive quiz generation system analyzing FSRS metrics to create personalized assessments from 1,400+ vocabulary words
- Designed **SQLite** database schema with 8+ tables supporting vocabulary content, user progress tracking, quiz analytics, and review history
- Published production application to iOS App Store with integrated audio pronunciations and user progress visualization

ChangePlusPlus

September 2024 – May 2025

Software Engineer

Nashville, TN

- Developing a full-stack mobile application for School the World (NPO) with 10-member Scrum team
- Built the application using **Expo** and **React Native**, authoring and executing 30+ **Jest** unit tests to achieve a 20% increase in code coverage
- Implemented database collection for school visits using **SQLite** and **TypeScript** to ensure type safety

Vanderbilt Data Science Club

November 2023 – May 2024

Software Developer

Nashville, TN

- Collaborated with ThriftSmart, the largest thrift store in Nashville, using Agile project management
- Built an image-based price appraiser using **Meta SAM** and **Google Lens API**, improving pricing accuracy for thrift store items
- Designed a **React** web application to streamline inventory management, reducing manual tracking efforts

PROJECTS

Country Guesser | *MongoDB, Express, React, Node.js, TypeScript*

August 2024 – November 2024

- Developed a geography trivia web application with an interactive map built on Leaflet
- Integrated REST Countries API and a reverse geocoding API to validate user input based on map interactions
- Designed a NoSQL database to securely store user credentials with JWT for authentication and bcrypt.js for password hashing

ChatApp | *NextJS, React, TailwindCSS, Socket.io, Prisma, SQL*

June 2023 – December 2023

- Developed a full-stack real-time chat application using WebSockets for low latency messaging
- Implemented a scalable backend using Prisma and MySQL, supporting efficient CRUD operations for messages and channels.
- Integrated NextAuth and OAuth to ensure secure user authentication and authorization

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, HTML/CSS, TypeScript, SQL

Frameworks/Libraries: ReactJS, NodeJS, Tailwind, MaterialUI, TensorFlow, NumPy, pandas, matplotlib, scikit-learn

Development Tools: VS Code, Docker, Git, npm, IntelliJ, CLion, Postman, Google Suite, Microsoft Office, Excel